MLS Salary, Roster Construction, and Team Performance

Dominic Hoar-Weiler (dhoarwei) Brendan Flynn (bflynn)
Nick Kourakis (nkouraki)

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1 Introduction

Major League Soccer operates under a complex roster-building system, yet teams vary widely in on-field performance.

1.1 Problem Statement

MLS teams operate under strict roster and salary constraints, yet there is no clear empirical consensus on which roster construction strategies lead to success.

1.2 Why This Matters

Understanding how teams should allocate limited salary budget is critical for competitive strategy in a capped league like MLS.

1.3 Research Question

How does salary allocation and roster construction relate to team performance in MLS?

More specifically, we study whether team payroll, salary concentration, and positional spending patterns are associated with outcomes such as expected goal differential, goals added, or points per game.

We use publicly available MLS salary and performance data to evaluate multiple statistical models. Each individual model addresses the same project question from a different angle, allowing us to compare which modeling framework is most informative for understanding roster construction strategy.

1.4 Preview of Results

Team success in MLS is not determined by how much teams spend, but by how efficiently they convert salary into player performance.

2 Data

We combine MLS salary information with team- and player-level performance data. • Salary data: TODO • Team performance data: TODO • Player performance data: TODO • Seasons included: TODO

2.1 Unit of analysis

We use two related units of analysis depending on the model: • Team-season for team-level models • Player-season for player-level models

2.2 Variables

2.2.0.1 Outcome Variables Potential response variables include: • Team expected goal differential (xGD) • Team goals added differential (g+ diff) • Points per game • Player goals added (g+) • Minutes-adjusted player contribution

2.2.0.2 Predictors Potential explanatory variables include: • Total payroll • Log payroll • Share of payroll spent on top 3 players • Salary concentration index (Gini / HHI) • Positional spending shares • Average age / experience (if available) • Designated player indicators (if available)

2.3 Data Cleaning Overview

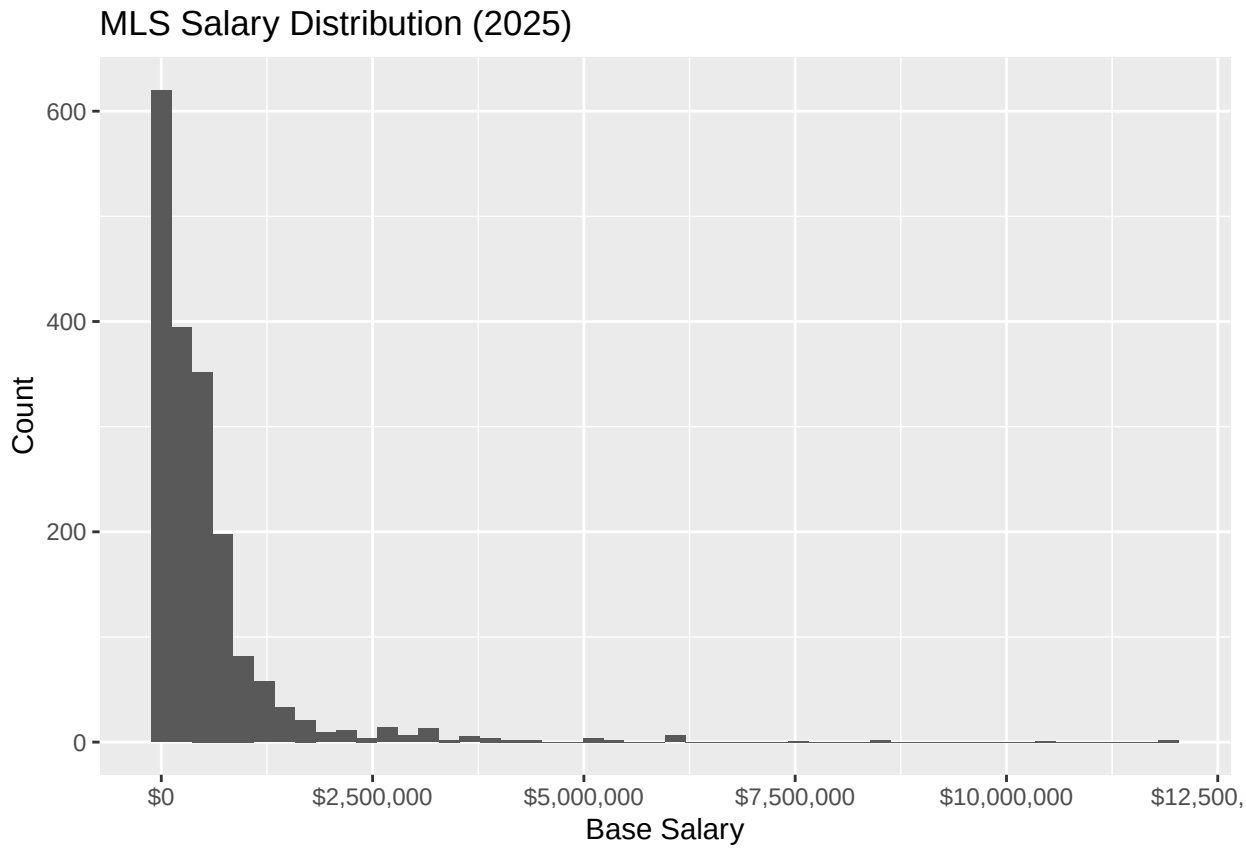
We construct a team-level dataset by aggregating player-level salary and performance data. Key steps include:

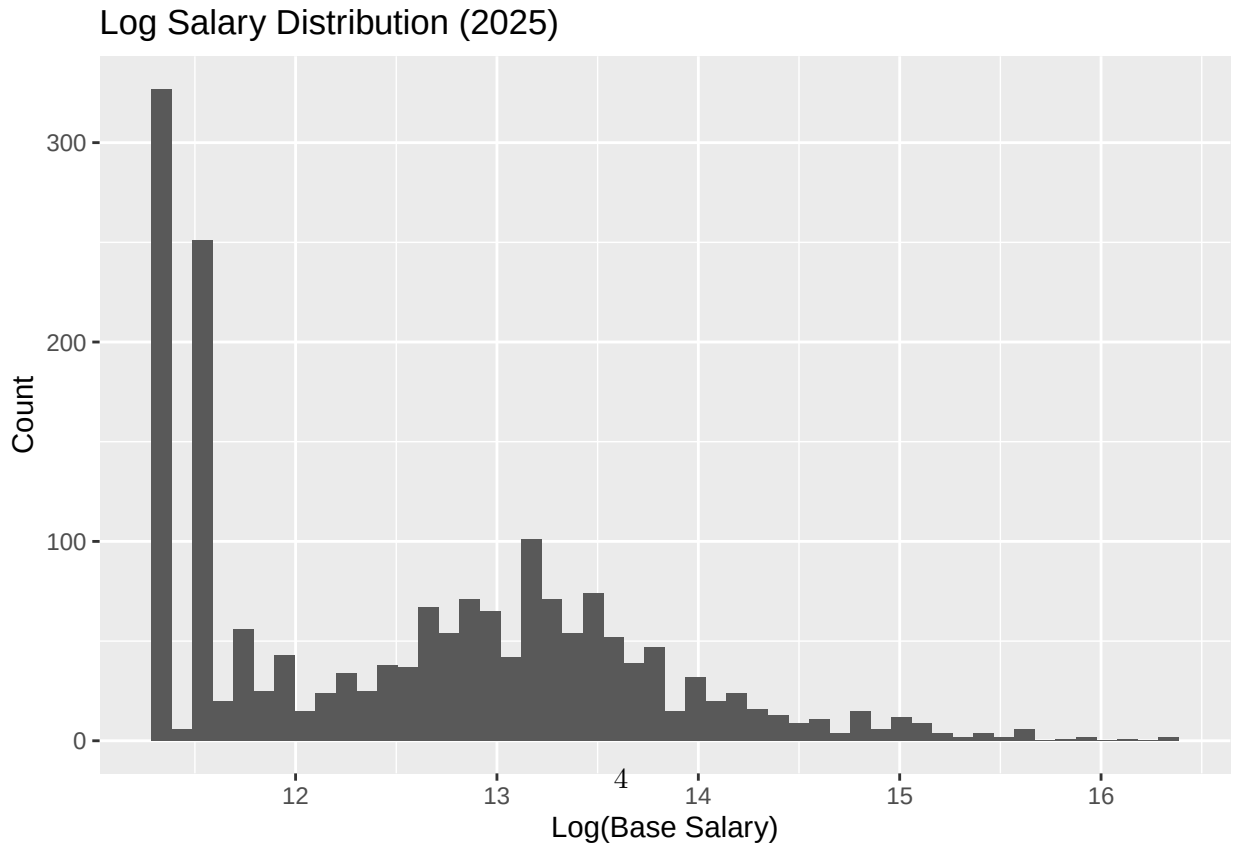
- Aggregating goals added into team-level performance
- Computing salary concentration (top 3 share)
- Creating proxies for designated players using top salary percentiles
- Constructing age-based U22 indicators
- Engineering features capturing roster balance (variance, low-salary share)

These transformations allow us to map individual salary structures to team-level outcomes.

3 Exploratory Data Analysis (EDA)

3.0.1 Salary distribution



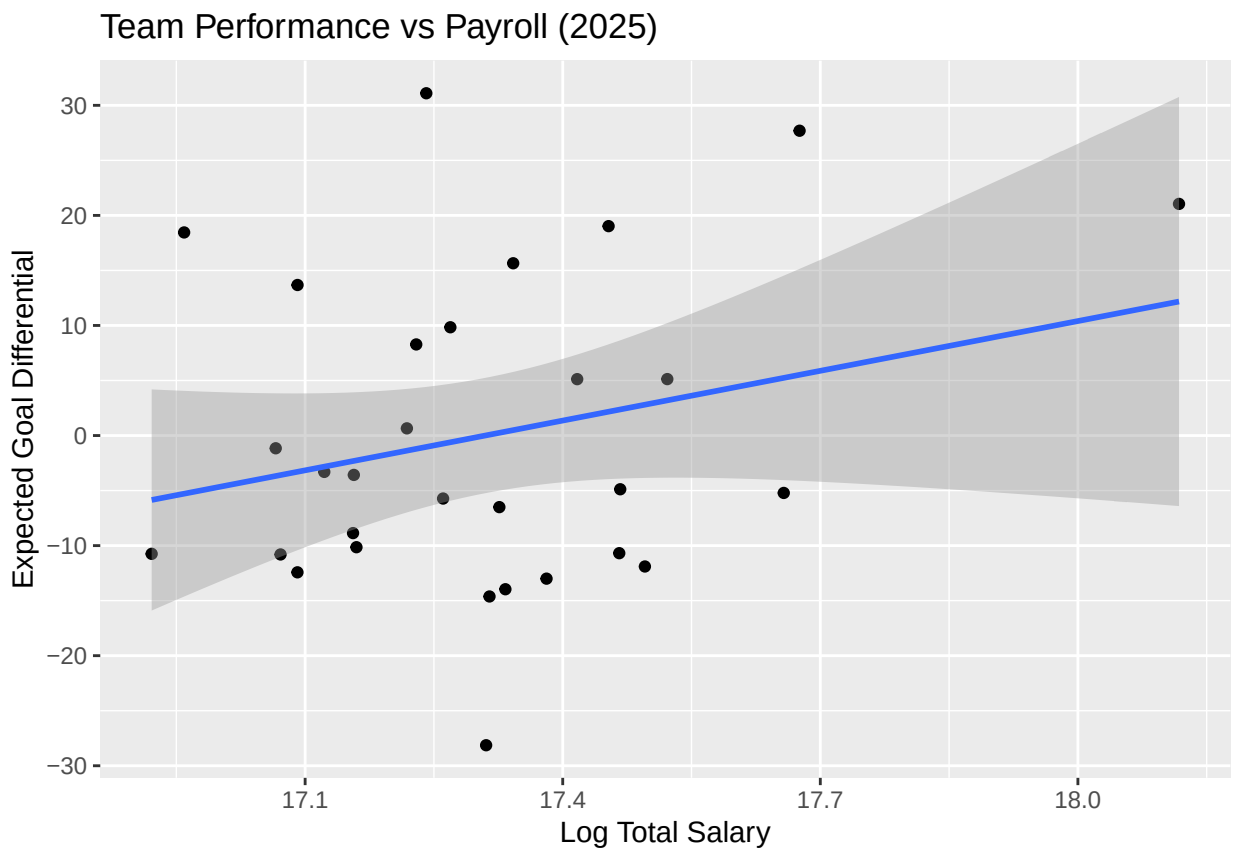


3.0.1.1 Takeaway

Salaries are highly right-skewed, meaning a small number of players earn disproportionately high salaries and suggesting a star-driven league structure.

3.0.2 Team Payroll vs Team Performance

```
## `geom_smooth()` using formula = 'y ~ x'
```

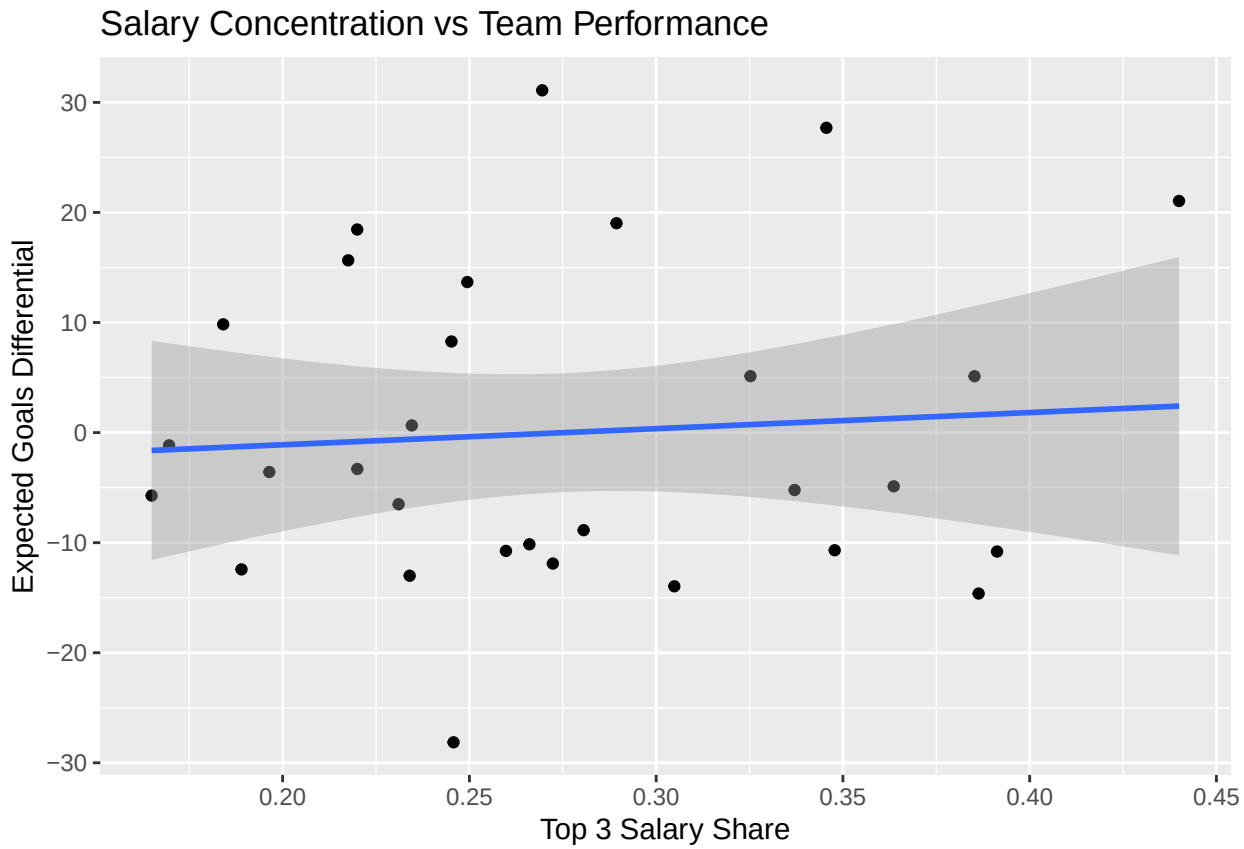


3.0.2.1 Takeaway

We see a weak/moderate positive relationship (likely). Payroll alone does not fully explain performance, which motivates out decision to add roster structure variables.

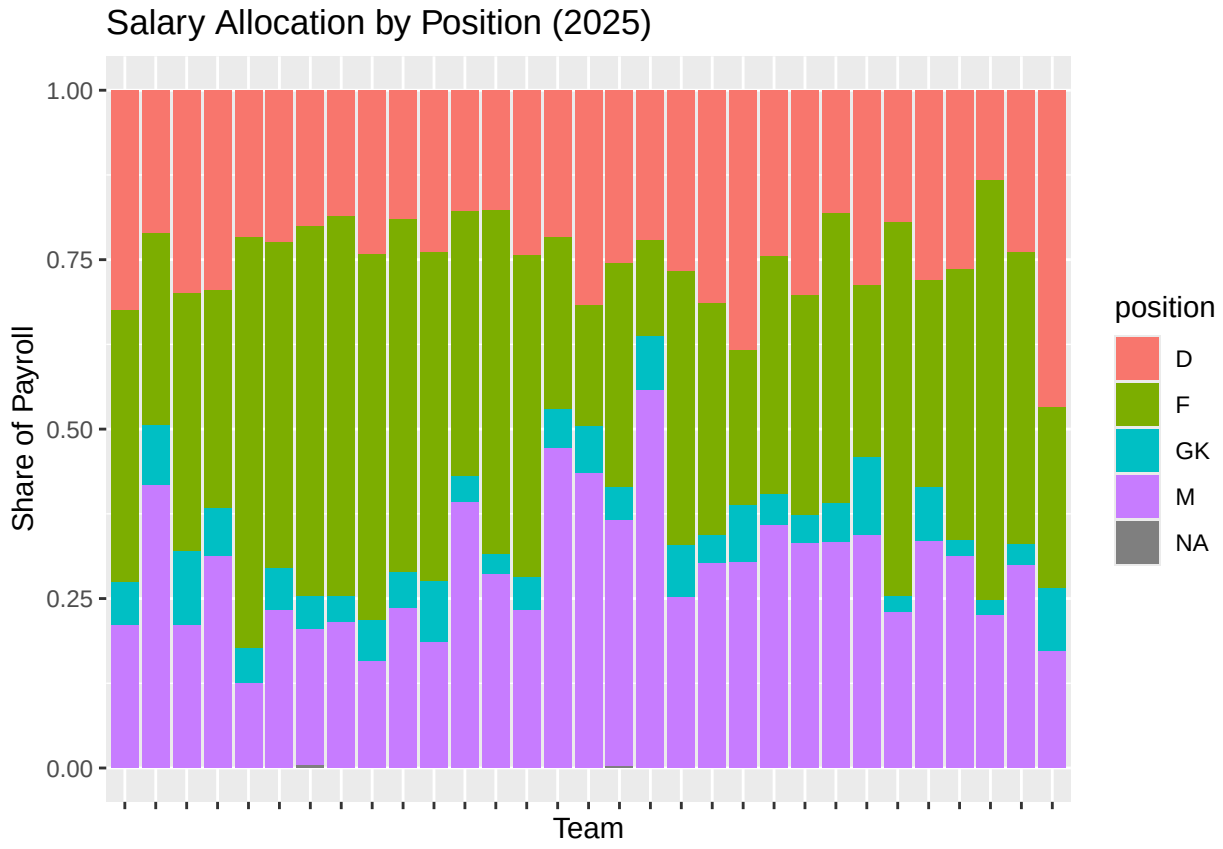
3.0.3 Salary concentration vs team performance

```
## `geom_smooth()` using formula = 'y ~ x'
```

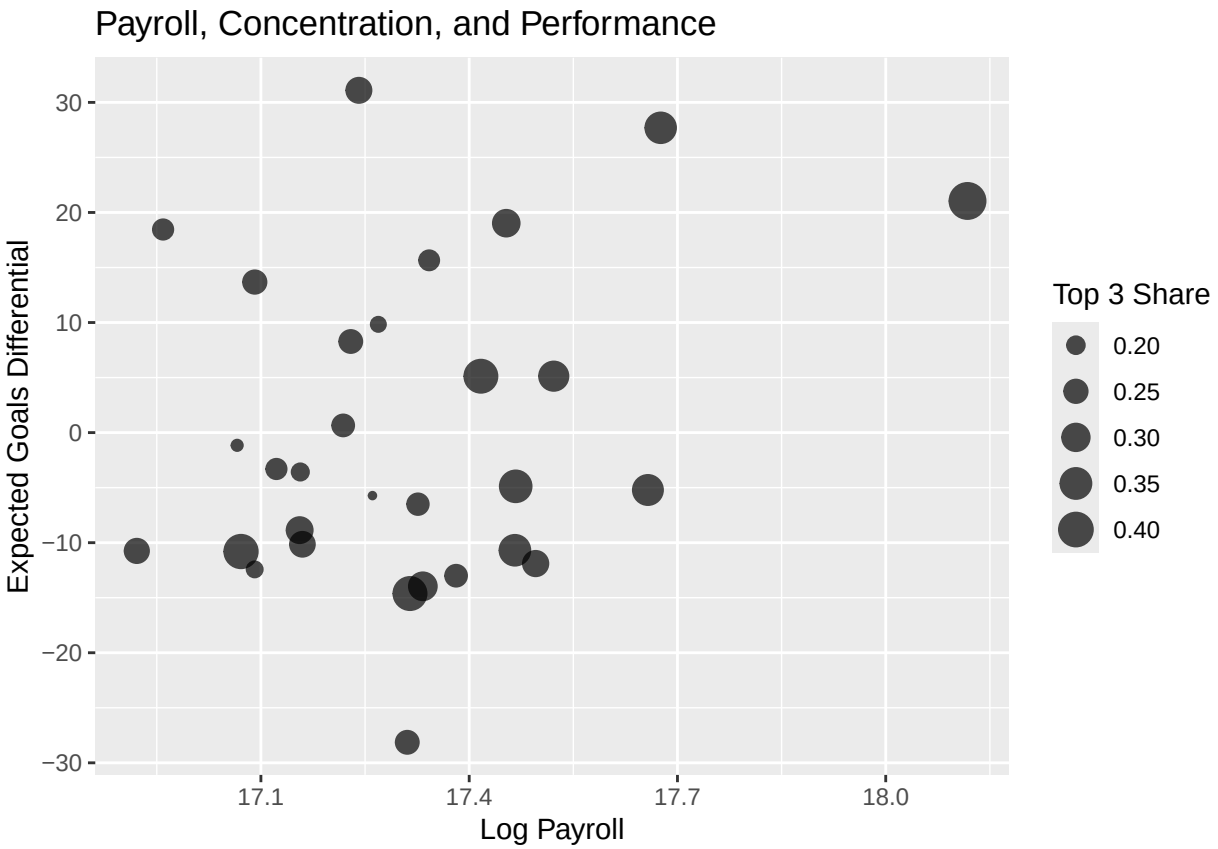


3.0.3.1 Takeaway The regression line is essentially flat. Confidence band is wide. Points are highly scattered. No visible monotonic trend.

3.0.4 Positional Allocation Patterns



EDA takeaway: TODO


Payroll, Concentration, and Performance
Expected Goals Differential
30
20
10
0
−10
−20
−30
17.1
17.4
17.7
18.0
Log Payroll
Top 3 Share
0.20
0.25
0.30
0.35
0.40

4 Methods

Across all individual analyses, the goal is to evaluate how roster construction and salary allocation relate to team success. Each group member fits a distinct model, motivated by a different statistical perspective.

4.1 Individual Analysis 1: Team-level linear regression (Dominic Hoar-Weiler)

4.1.1 Question

Does team performance increase with total payroll, and does salary concentration help or hurt after controlling for total spending?

4.1.2 Model specification

4.1.2.1 Baseline Model

$$xGD_{it} = \beta_0 + \beta_1 \log(\text{Salary}_{it}) + \gamma_t + \varepsilon it$$

4.1.2.2 Player Level GAM

$$GAAR_{ijt} = f(\log(\text{Salary}_{ijt})) + \delta pos(j) + f_{pos(j)}(\log(\text{Salary}_{ijt})) + \varepsilon_{ijt}$$

4.1.2.3 Team Level Construction

$$\widehat{V}it = \sum_{j \in i,t} \widehat{GAAR}_{ijt} \cdot \frac{\text{Minutes}_{ijt}}{90}$$

4.1.2.4 Team Efficiency Model

$$xGD_{it} = \beta_0 + \beta_1 \widehat{V}_{it} + \beta_2 \text{Salary}_{it} + \beta_3 \text{Age}_{it} + \beta_4 \text{PctAtt}_{it} + \beta_5 \text{PctMid}_{it} + \gamma_t + \varepsilon_{it}$$

4.1.2.5 Team Multilevel Model

$$\begin{aligned} xGD_{it} = \beta_0 &+ \beta_1 \,\text{Value}_{it} + \beta_2 \,\text{Salary}_{it} + \beta_3 \,\text{Age}_{it} \\ &+ \beta_4 \,\text{Top3Share}_{it} + \beta_5 \,\text{U22}_{it} \\ &+ \beta_6 \,\text{ForwardShare}_{it} + \beta_7 \,\text{MidfieldShare}_{it} \\ &+ u_i + \epsilon_{it} \end{aligned}$$

4.1.3 Assumptions

- `Linearity`
- `Independence across team-seasons`
- `Constant variance`
- `Limited omitted variable bias`

4.1.4 Why this model is appropriate

This provides a simple, interpretable baseline model linking roster spending strategy to performance.

4.1.5 Model evaluation plan

- `RMSE / MAE`
- `Residual diagnostics`
- `Compare alternative predictor sets`

4.1.6 Uncertainty plan

- `95% confidence intervals for coefficients`
- `Optional bootstrap confidence intervals`

4.1.7 SIDE NOTES DOM

When pooling all seasons without controlling for year effects, salary variables are not statistically significant. However, once season fixed effects are introduced, total payroll becomes a significant predictor of team performance. This suggests that cross-season structural changes in MLS—such as salary cap growth and evolving roster rules—obscure the relationship between spending and performance when not properly controlled for.

4.2 Individual Analysis 2: Roster Architecture (Nick Kourakis)

4.2.1 Question

Does team performance depend more on the salary disparity between highest and lowest paid starters, or on the the depth of it's TAM-eligible[1] roster?

4.2.2 Model specification

4.2.2.1 Baseline Model

$$xGD_{it} = \beta_0 + \beta_1 Gini_{it} + \varepsilon_{it}$$

4.2.2.2 Dispersion Model

$$xGD_{it} = \beta_0 + \beta_1 \text{Gini}_{it} + \beta_2 \text{MidClassDepth}_{it} + \beta_3 \log(\text{TotalSalary}_{it}) + \varepsilon_{it}$$

Evaluates the importance of TAM players, controlling for total team spending.

4.2.2.3 Multi Season Roster Model

$$xGD_{it} = \beta_1 \text{Gini}_{it} + \beta_2 \text{MidClassDepth}_{it} + \alpha_i + \gamma_t + \varepsilon_{it}$$

Within a specific team over several years, does performance improve when that specific team moves toward a more balanced roster?

4.2.3 Assumptions

- `Linearity`
- `No Multicollinearity`
- `Independence`

4.2.4 Why this model is appropriate

Due to the MLS salary cap, every dollar spent on a "Star" is a dollar that could've been used somewhere else on the roster. The previous analysis evaluates spending efficiency, this model tests the roster structure. By using the Gini coefficient and TAM-depth, we can determine if success comes from a roster that is top-heavy (high inequality) or from a deep roster (high middle-class density).

[1] TAM (Targeted Allocation Money) Players: The MLS "middle class," consisting of high-level starters earning above the Maximum Salary Budget Charge ($803,125) but below the Designated Player threshold. These players' budget hits are "bought down" using league-allocated funds. See the official documentation here.

4.2.5 Model evaluation plan

- Adjusted R^2: To measure how much additional variance in xGD is explained by roster distribution compared to a baseline model.
- Akaike Information Criterion (AIC): To determine if adding "Middle Class" complexity improves model fit without over-fitting.
- Coefficient Analysis: Evaluating the significance of β_1 (Gini) to identify whether inequality is correlated with performance.

4.2.6 Uncertainty plan

- 95% confidence intervals: To verify that the effects of Gini and TAM-depth are statistically distinguishable from zero.
- Clustered Standard Errors: Grouped by team_id to account for correlation in performance across multiple seasons.
- Sensitivity Analysis: Testing the model with and without extreme salary outliers.

4.3 Individual Analysis 3 (Brendan Flynn)

4.3.1 Question

4.3.2 Model specification

4.3.3 Assumptions

4.3.4 Why this model is appropriate

4.3.5 Model evaluation plan

4.3.6 Uncertainty plan

5 Results

5.1 Individual Analysis 1 Results (Dom)

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
## See help("Deprecated")

## Warning in attr(x, "format"): 'xfun::attr()' is deprecated.
## Use 'xfun::attr2()' instead.
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```

Table 1: Baseline Team Model

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-84.067	32.847	-2.559	0.011	-148.705	-19.428
log_salary	5.159	2.009	2.568	0.011	1.205	9.113
season_name2014	1.063	3.779	0.281	0.779	-6.375	8.500
season_name2015	-0.276	3.711	-0.074	0.941	-7.578	7.027
season_name2016	-0.914	3.726	-0.245	0.806	-8.247	6.419
season_name2017	-1.442	3.669	-0.393	0.695	-8.663	5.779
season_name2018	-2.258	3.696	-0.611	0.542	-9.530	5.015
season_name2019	-2.778	3.717	-0.748	0.455	-10.092	4.535
season_name2020	0.582	3.502	0.166	0.868	-6.309	7.474
season_name2021	-3.329	3.702	-0.899	0.369	-10.613	3.956
season_name2022	-3.978	3.774	-1.054	0.293	-11.405	3.449
season_name2023	-4.424	3.827	-1.156	0.249	-11.955	3.107
season_name2024	-4.835	3.902	-1.239	0.216	-12.513	2.843
season_name2025	-5.236	3.960	-1.322	0.187	-13.029	2.557

```
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```

Table 2: Main Team Efficiency Model

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	46.246	24.075	1.921	0.056	-1.284	93.775
predicted_team_value	0.085	0.011	7.507	0.000	0.063	0.107
player_total_salary	0.000	0.000	-3.522	0.001	0.000	0.000
factor(season_name)2014	-7.126	3.858	-1.847	0.067	-14.743	0.491

term	estimate	std.error	statistic	p.value	conf.low	conf.high
factor(season_name)2015	-7.253	4.543	-1.596	0.112	-16.222	1.716
factor(season_name)2016	-10.302	3.871	-2.662	0.009	-17.944	-2.660
factor(season_name)2017	-13.651	4.032	-3.385	0.001	-21.612	-5.690
factor(season_name)2018	-18.978	4.152	-4.571	0.000	-27.175	-10.781
factor(season_name)2019	-23.790	5.049	-4.712	0.000	-33.757	-13.822
factor(season_name)2020	0.396	5.057	0.078	0.938	-9.587	10.379
factor(season_name)2021	-21.427	4.286	-4.999	0.000	-29.890	-12.965
factor(season_name)2022	-22.113	4.672	-4.734	0.000	-31.336	-12.890
factor(season_name)2023	-27.918	5.684	-4.912	0.000	-39.140	-16.696
factor(season_name)2024	-25.560	5.395	-4.738	0.000	-36.212	-14.909
factor(season_name)2025	-32.808	6.021	-5.449	0.000	-44.694	-20.922
avg_age	-2.553	0.856	-2.984	0.003	-4.242	-0.864
pct_att	-24.611	18.464	-1.333	0.184	-61.065	11.842
pct_midfielders	-15.622	15.293	-1.022	0.308	-45.814	14.570

```
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```

Table 3: Player-Level GAM: Parametric Terms

term	estimate	std.error	statistic	p.value
(Intercept)	1.422	0.033	43.724	0
positionF	0.452	0.050	8.992	0
positionGK	0.000	0.000	NaN	NaN
positionM	0.404	0.045	8.991	0

```
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## See help("Deprecated")
```

Table 4: Player-Level GAM: Smooth Terms

term	edf	ref.df	statistic	p.value
s(log_salary)	3.187	3.673	44.933	0.000
s(log_salary):positionD	1.000	1.000	23.397	0.000

term	edf	ref.df	statistic	p.value
s(log_salary):positionF	1.843	1.971	8.242	0.000
s(log_salary):positionGK	1.000	1.000	0.373	0.541
s(log_salary):positionM	1.014	1.025	9.718	0.002

```
## Warning in attr(x, "align"): 'xfun::attr()' is deprecated.
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## See help("Deprecated")
```

Table 5: Extended Multilevel Model: Fixed Effects

effect	term	estimate	std.error	statistic	df	p.value	conf.low	conf.high
fixed	(Intercept)	-11.013	6.009	-1.833	307.500	0.068	-22.837	0.811
fixed	pred_val_s	6.495	0.983	6.608	304.342	0.000	4.561	8.429
fixed	salary_s	-6.092	1.161	-5.249	307.898	0.000	-8.376	-3.809
fixed	age_s	-1.169	0.708	-1.651	303.605	0.100	-2.562	0.225
fixed	top3_share	14.216	7.214	1.971	307.905	0.050	0.021	28.412
fixed	u22_proxy_count	-0.319	0.565	-0.565	306.824	0.572	-1.430	0.792
fixed	share_F	5.922	9.692	0.611	305.422	0.542	-13.149	24.992
fixed	share_M	7.223	9.204	0.785	299.909	0.433	-10.891	25.336

```
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## See help("Deprecated")
```

Table 6: Extended Multilevel Model: Random Effects

effect	group	term	estimate	conf.low	conf.high
ran_pars	team_id	sd__(Intercept)	5.074	NA	NA
ran_pars	Residual	sd__Observation	9.583	NA	NA

5.1.0.1 Baseline Model We first estimated a baseline linear regression model relating team performance (goals added differential) to total payroll and salary concentration (top-three salary share). The model explains very little variation in team performance ($R^2 \approx 0.02$), and neither predictor is statistically significant. While the coefficient on log payroll is positive, the estimate is highly uncertain, and salary concentration also shows no clear relationship with performance.

Overall, these results suggest that simple measures of payroll and salary concentration alone do not adequately explain variation in team performance in MLS.

5.1.0.2 Player Level GAM At the player level, salary clearly translates into performance, but not in a simple linear way. The GAM shows strong nonlinear effects and meaningful differences across positions, with attackers and midfielders generating more value on average and exhibiting different salary-to-performance curves. This suggests that while spending more does generally increase individual performance, the returns are heterogeneous and exhibit diminishing effects depending on how salary is allocated across positions.

5.1.0.3 Team Model LM At the team level, once player contributions are aggregated using minutes-weighted predicted value, performance becomes substantially more predictable. Teams with higher total player value perform significantly better, while total salary—holding value constant—has a negative and statistically significant effect. This indicates that spending alone does not drive success; rather, performance depends on how efficiently teams convert payroll into on-field production.

Adding team-level controls such as age and roster composition does not change this core result. Predicted player value remains strongly positive and significant, and salary continues to have a negative conditional effect, reinforcing the idea that inefficiency in spending is detrimental to performance.

5.1.0.4 Multilevel Team Model Additional results provide further context. Team age has a negative and significant effect, suggesting older teams tend to perform worse even after controlling for value and spending, while roster composition variables do not have strong independent effects. A multilevel model incorporating team and season random effects confirms substantial unobserved heterogeneity, but the core findings remain unchanged: player value is strongly positively associated with performance, and salary is negatively associated once value is accounted for. This demonstrates that the main results are robust to differences across teams and seasons.

Then interpretation: • $\beta_1 > 0$: higher player value improves performance • $\beta_2 < 0$: overpaying reduces performance • $\beta_4 > 0$: top-heavy salary structures improve outcomes • $\beta_5 \approx 0$: no clear U22 effect • $\beta_6, \beta_7 \approx 0$: no clear positional allocation effect

Emphasize (these are strong)

- Value vs salary (efficiency)
- Top3 concentration effect

Neutral / exploratory

- Age

Downplay

- U22 proxy (data limitation)
- Positional shares

5.1.0.5 Answer to Research Question Across all modeling approaches, we find that team performance in MLS is not driven simply by total payroll or salary concentration, but rather by how effectively teams translate spending into on-field production.

The baseline linear model shows that total payroll and salary concentration alone have weak explanatory power for team performance, with low R^2 and statistically insignificant coefficients. This suggests that aggregate spending levels do not meaningfully differentiate successful and unsuccessful teams.

At the player level, the generalized additive model (GAM) reveals a strong but nonlinear relationship between salary and performance, with substantial heterogeneity across positions. This indicates that while higher salaries are generally associated with better individual performance, the marginal returns to spending vary depending on how salary is allocated across players and positions.

When aggregating player-level predicted value to the team level, team performance becomes highly predictable. The team-level model shows that predicted player value is strongly positively associated with performance, while total salary, conditional on value, has a negative and statistically significant effect. This implies that inefficiency in spending—paying more for the same level of production—is detrimental to team success.

Finally, the multilevel model confirms that these relationships are robust to unobserved heterogeneity across teams and seasons. The positive effect of player value and the negative conditional effect of salary persist even after accounting for random effects.

Taken together, these results suggest that roster construction strategy in MLS is best understood through the lens of efficiency rather than scale. Successful teams are not necessarily those that spend the most, but those that allocate salary in a way that maximizes player value per dollar.

5.2 Individual Analysis 2 Results (Nick)

5.3 Individual Analysis 3 Results (Flynn)

6 Discussion

Our project examined whether MLS teams benefit more from spending heavily overall, concentrating spending in a few stars, or distributing spending more efficiently across the roster.

7 Appendix

7.1 Section A: Code

Describe any major cleaning steps here